

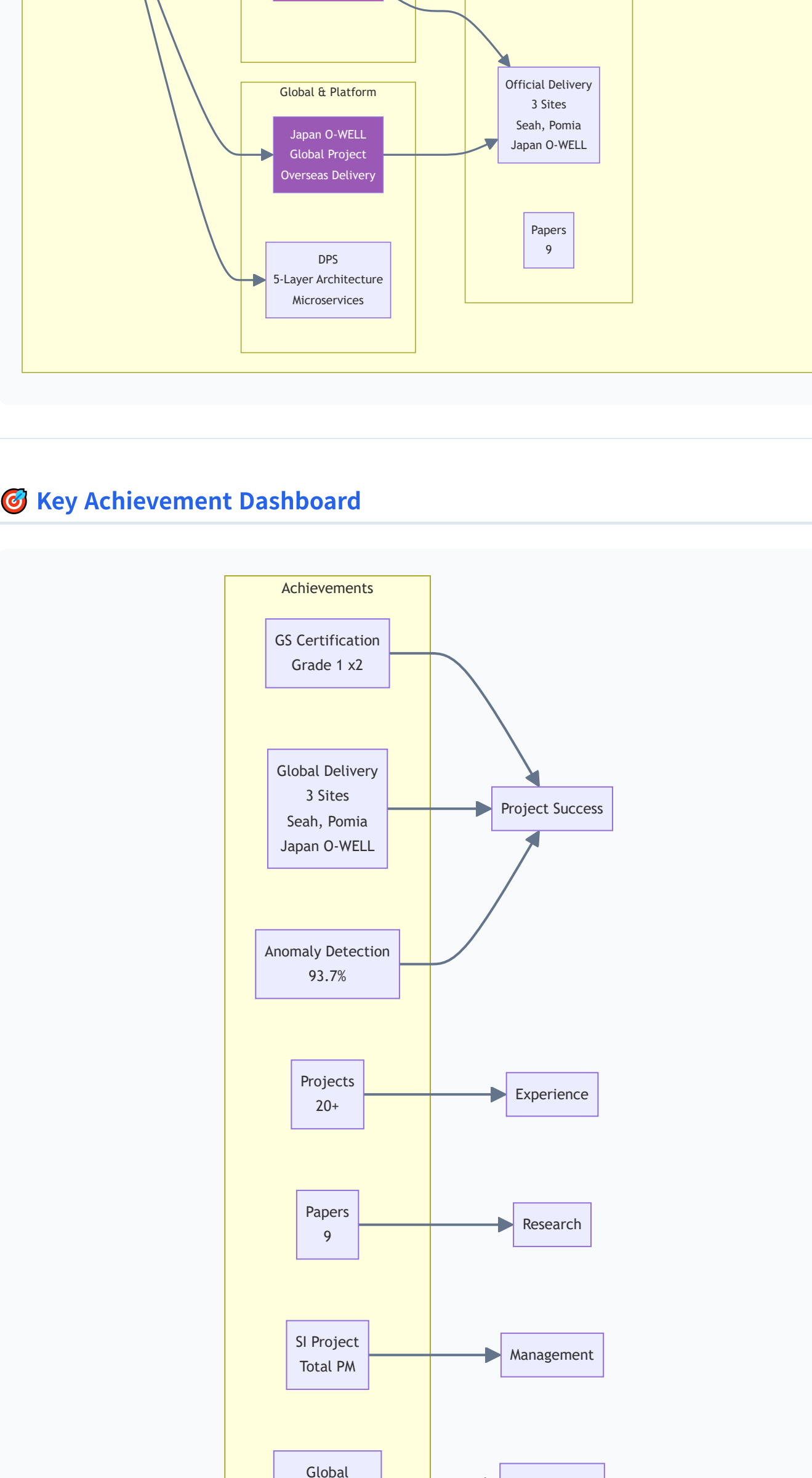
Kwon Sunryong - Portfolio

"A field-oriented researcher who organizes knowledge structures, prioritizing information over data, and data over models"

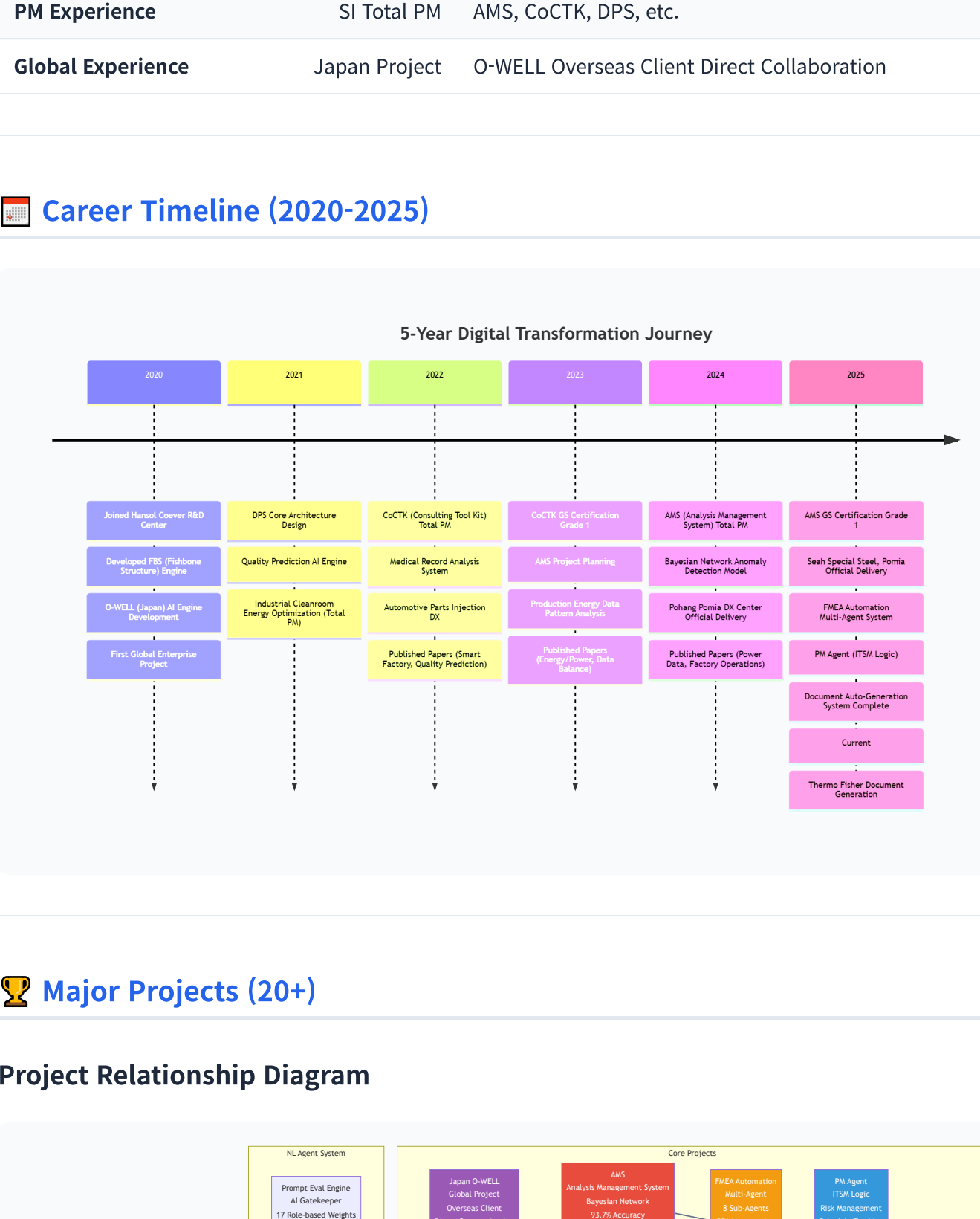
Basic Information

Name: Kwon Sunryong (권순룡)
GitHub: <https://github.com/moobaek>

Portfolio Structure (Overview)



Key Achievement Dashboard



Category	Metric	Details
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Certification	GS Grade 1 x2	CoCTK (2024), AMS (2025)
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Global Delivery	3 Sites	Seah Special Steel, Pomia, Japan O-WELL
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Accuracy	93.7%	Bayesian Network-based Anomaly Detection
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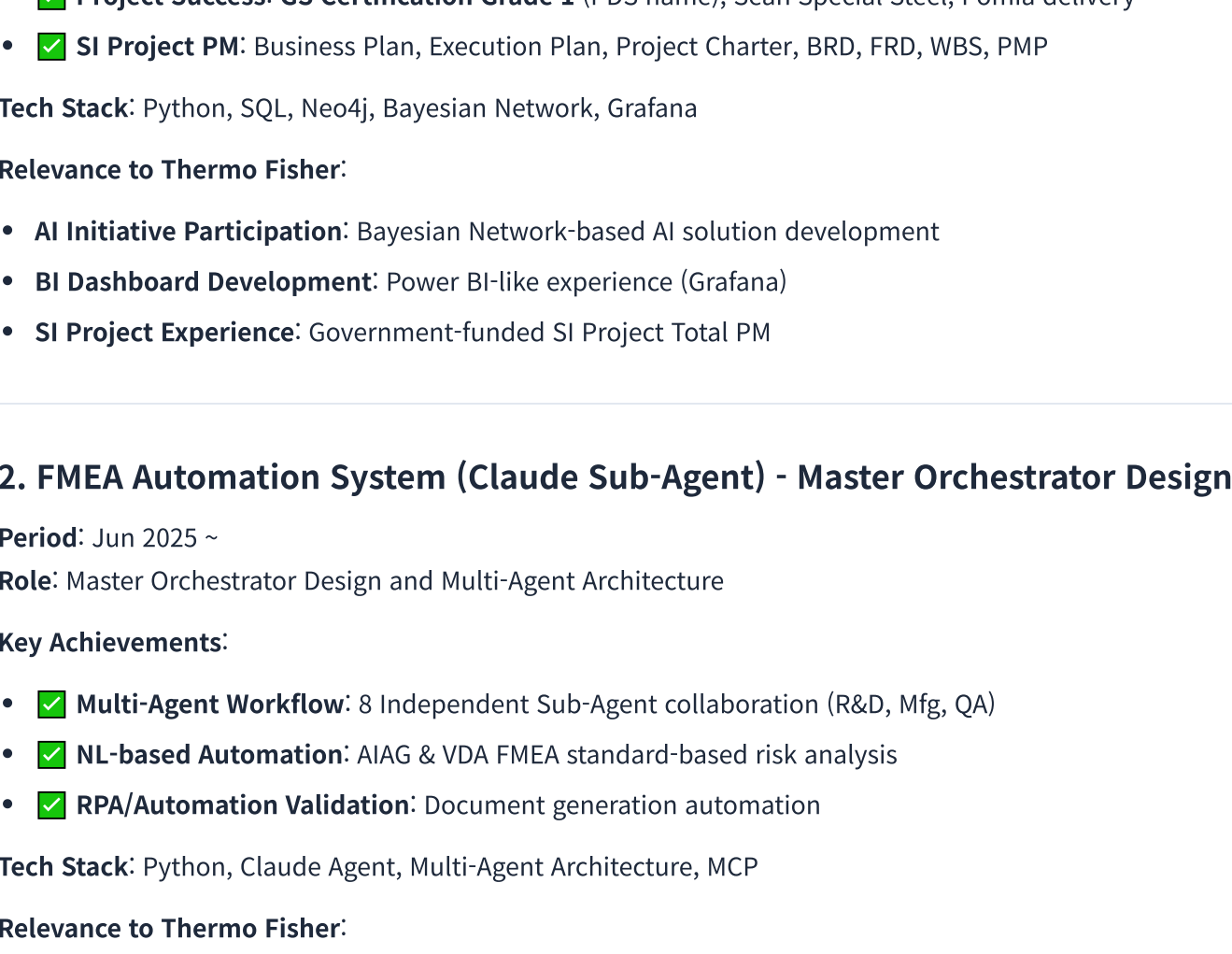
Projects	20+	AI, Platform, Sensors, Energy Optimization
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Papers	9	Academic papers published 2022-2025
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PM Experience	SI Total PM	AMS, CoCTK, DPS, etc.
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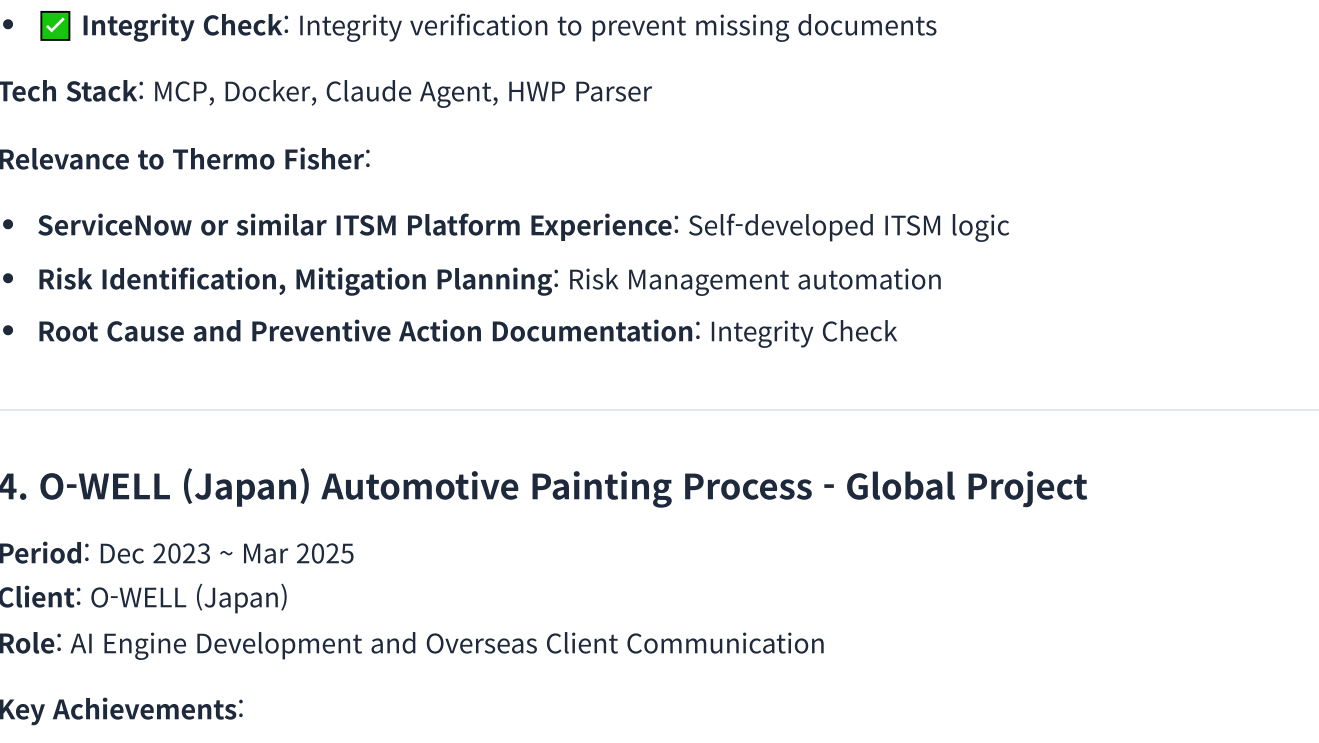
Global Experience	Japan Project	O-WELL Overseas Client Direct Collaboration
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Career Timeline (2020-2025)



Major Projects (20+)

Project Relationship Diagram



1. AMS (Analysis Management System) - Total PM

Period: Jul 2024 ~ Mar 2025
Client: Korea Institute for Advancement of Technology (KIAT)
Role: Total PM for AI Integrated Platform Development

Key Achievements:

- ✅ **Bayesian Network-based Anomaly Detection Model:** Probability optimization (gradient descent), 93.7% accuracy
- ✅ **Grafana Real-time Dashboard:** Anomaly detection visualization (Power BI-like experience)
- ✅ **Project Success: GS Certification Grade 1** (PDS name), Seah Special Steel, Pomia delivery
- ✅ **SI Project PM:** Business Plan, Execution Plan, Project Charter, BRD, FRD, WBS, PMP

Tech Stack: Python, SQL, Neo4j, Bayesian Network, Grafana

Relevance to Thermo Fisher:

- AI Initiative Participation: Bayesian Network-based AI solution development
- BI Dashboard Development: Power BI-like experience (Grafana)
- SI Project Experience: Government-funded SI Project Total PM

2. FMEA Automation System (Claude Sub-Agent) - Master Orchestrator Design

Period: Jun 2025 ~
Role: Master Orchestrator Design and Multi-Agent Architecture

Key Achievements:

- ✅ **Multi-Agent Workflow:** 8 Independent Sub-Agent collaboration (R&D, Mfg, QA)
- ✅ **NL-based Automation:** AIAG & VDA FMEA standard-based risk analysis
- ✅ **RPA/Automation Validation:** Document generation automation

Tech Stack: Python, Claude Agent, Multi-Agent Architecture, MCP

Relevance to Thermo Fisher:

- AI Initiatives: Use case identification, pilots, AI solutions to business workflows
- RPA Solution Experience: Workflow automation, document automation
- AI Adoption Best Practices Documentation: FMEA automation process documentation

3. PM Agent (Business Management Sub-Agent) - ITSM Logic Implementation

Period: Oct 2025 ~
Role: System Design and Development Lead

Key Achievements:

- ✅ **Self-developed ITSM Logic:** ServiceNow-like functionality
- ✅ **Risk Management:** Automatic toxic clause extraction and risk assessment
- ✅ **Schedule Tracking:** Automatic timeline updates through meeting minutes analysis
- ✅ **Integrity Check:** Integrity verification to prevent missing documents

Tech Stack: MCP, Docker, Claude Agent, HWP Parser

Relevance to Thermo Fisher:

- ServiceNow or similar ITSM Platform Experience: Self-developed ITSM logic
- Risk Identification, Mitigation Planning: Risk Management automation
- Root Cause and Preventive Action Documentation: Integrity Check

4. O-WELL (Japan) Automotive Painting Process - Global Project

Period: Dec 2023 ~ Mar 2025
Client: O-WELL (Japan)
Role: AI Engine Development and Overseas Client Communication

Key Achievements:

- ✅ **Global Collaboration:** Direct communication with Japanese client
- ✅ **Causal Relationship Diagram AI Engine:** Pattern optimization + hierarchical clustering
- ✅ **Company-wide DX Acceleration:** Completion reports and deliverables

Tech Stack: Python, AI Engine, Data Analysis

Relevance to Thermo Fisher:

- Global or Multinational Environment Experience: Japan global enterprise project
- Business-level English Proficiency: Overseas client communication
- Internal/External Stakeholder Collaboration: Technical discussions, requirements analysis

5. DPS (Data Collection System) - PM

Period: 2021 ~ 2024
Role: Core Architecture Design and Development (PM)

Key Achievements:

- ✅ **5-Layer Architecture Design:** Microservices, Neo4j Graph DB
- ✅ **Large-scale Data Processing:** Metal industry 5 major process AI automation
- ✅ **Docker/Kubernetes Infrastructure:** Container orchestration

Tech Stack: Python, SQL, Neo4j, Microservices, Docker, Kubernetes

Relevance to Thermo Fisher:

- IT Operations, Application Support, Digital Systems Management: 5 years experience
- Structured Approach to Multiple Priorities: 5-layer architecture design and operation

6. Document Auto-Generation System (NL Agent)

Period: 2024 ~ Present
Role: System Design and Development Lead

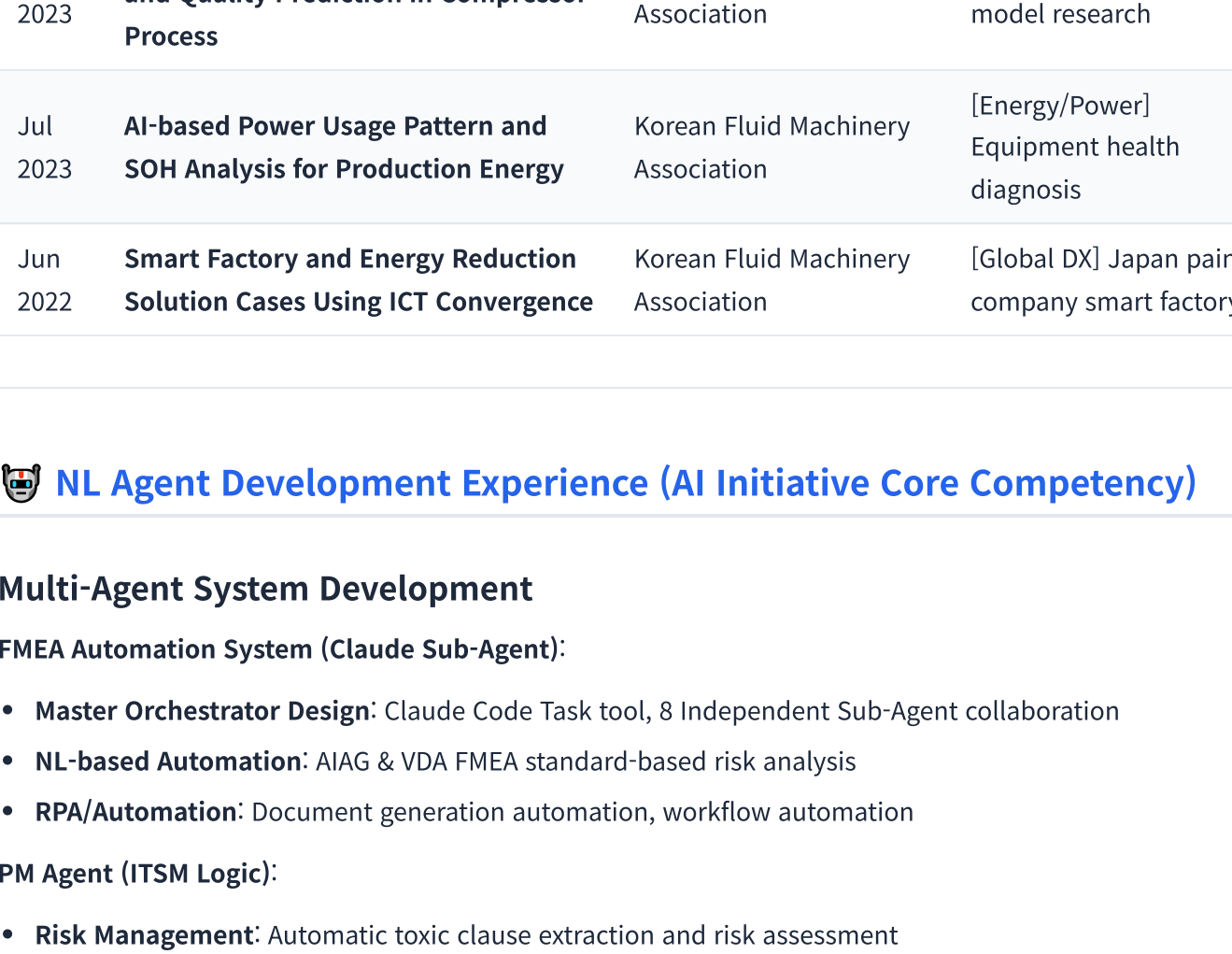
Key Achievements:

- ✅ **NL Agent-based Automation:** Claude Sub-Agent Multi-Agent Workflow
- ✅ **Customized Document Generation:** Resume and portfolio matching job requirements
- ✅ **Smart Matching System:** AI auto-selects relevant projects and experience
- ✅ **Actual Application:** This document (Thermo Fisher Portfolio) generated by this system

Tech Stack: Claude Sub-Agent, Multi-Agent Workflow, Python, MCP, Prompt Engineering

Document Auto-Generation System Evolution

Development Roadmap



Generated Documents List

Generation Method	Project	Document Type	Period	Result
Auto	Thermo Fisher	Resume/Portfolio	Jan 2026	This Document
Auto	Hyundai Home Shopping	Resume/Portfolio/Cover Letter	Jan 2026	Customized Documents
Auto	Seah Special Steel	Execution Plan (SOW)	Jun 2025	5-month Project Success
Manual	AMS	Business Plan, BRD/FRD, WBS/PMP	Jul 2024	GS Grade 1, Delivery
Manual	Japan O-WELL	Completion Reports and Deliverables	Mar 2025	Global Enterprise Delivery

Tech Stack Map



Academic Achievements (9 Papers)

Date	Paper Title	Journal/Conference	Key Achievement
Dec 2025	FMEA Generation Research Based on Analysis Correlation/Probability Network Optimal Path	KSFM 2025 Winter Conference	[FMEA/AMS] FMEA auto-generation tech validation
Jun 2025	Structure-Probability Integrated Network and Optimal Management Using AI	Korean Fluid Machinery Association	[AMS] Fishbone AI model academic advancement
Dec 2024	Clustering Technique for Factory Operation Key Factor Identification	Korean Production Manufacturing Society	[Pomia DX] Multi-dimensional factory data analysis
Dec 2024	Equipment Abnormal State-based Optimal Process Data Inference	Korean Fluid Machinery Association	[AMS] Real-time risk management algorithm
Jul 2024	Equipment State Inference and Anomaly Prediction through Power Data	Korean Fluid Machinery Association	[Energy/Sensor] Power-based predictive maintenance
Dec 2023	Power Quality Analysis and Energy Reduction for Variable Load Blowers	Korean Fluid Machinery Association	[Energy] 20% energy reduction validation
Dec 2023	AI System for Data Balance Problem and Quality Prediction in Compressor Process	Korean Fluid Machinery Association	[AI/Data] AI learning model research
Jul 2023	AI-based Power Usage Pattern and SOH Analysis for Production Energy	Korean Fluid Machinery Association	[Energy/Power] Equipment health diagnosis
Jun 2022	Smart Factory and Energy Reduction Solution Cases Using ICT Convergence	Korean Fluid Machinery Association	[Global DX] Japan paint company smart factory

NL Agent Development Experience (AI Initiative Core Competency)

Multi-Agent System Development

FMEA Automation System (Claude Sub-Agent):

- Master Orchestrator Design: Claude Code Task tool, 8 Independent Sub-Agent collaboration
- NL-based Automation: AIAG & VDA FMEA standard-based risk analysis
- RPA/Automation: Document generation automation, workflow automation

PM Agent (ITSM Logic):

- Risk Management: Automatic toxic clause extraction and risk assessment
- Schedule Tracking: Automatic timeline updates through meeting minutes
- Integrity Check: Integrity verification to prevent missing documents

Prompt Evaluation Engine (Claude Sub-Agent):

- AI Gatekeeper: Controls all AI output entrance, full prompt evaluation
- 3 Core Dimension Evaluation: Quality, Consistency, Cost
- 17 Role-based Dynamic Weight System

Document Auto-Generation System:

- Job Posting Auto-Parsing: Requirements, tech stack auto-extraction
- Portfolio Smart Matching: AI auto-selects relevant projects
- Actual Application: This document (Thermo Fisher Portfolio) generated by this system

Tech Stack: Claude Sub-Agent, Multi-Agent Workflow, Python, MCP, Prompt Engineering, LangGraph, FastAPI

Links

GitHub

- Main Repository: https://github.com/moobaek/Testing_AI_agents_for_public_use
- Portfolio Documents: https://github.com/moobaek/Testing_AI_agents_for_public_use/tree/main/portfolio/portfolio_docs
- GitHub Profile: <https://github.com/moobaek>